ASVspoof 5: Voice Forgery Detection and Generalization to New Attacks

Wang et al. (2024), in their paper ASVspoof 5: Crowdsourced Speech Data, Deepfakes, and Adversarial Attacks at Scale, present the fifth edition of the ASVspoof challenge, which is intended for the development and evaluation of systems for detecting voice forgeries. The study focuses on the difficulty of detection systems in generalizing beyond their training conditions, especially when they are exposed to new speakers, to diverse acoustic conditions, and to advanced methods for generating synthetic speech.

The researchers developed a dataset based on the English speech data of Multilingual LibriSpeech (MLS), which originates in thousands of speakers and in diverse recording conditions. The dataset includes 32 attack IDs based on Text-to-Speech and Voice Conversion systems, alongside Adversarial attacks intended to mislead detection systems. In order to examine the generalization ability, a separation was defined between speakers and between forgery systems in the training, development and evaluation sets. In addition, various compression and audio processing conditions were examined, including the use of neural codecs.

In the stand-alone forgery detection track (Track 1), the researchers used RawNet2 and AASIST as Baseline models. The models achieved EER rates of 36.04% and 29.12%, respectively, which indicate a significant difficulty under the challenge conditions. In contrast, the leading system in the minDCF ranking under Closed conditions achieved an EER of 8.61%, and the leading system under Open conditions achieved an EER of 2.59%. The researchers note that many of the leading systems used pre-trained speech representations based on Self-Supervised Learning, such as wav2vec 2.0, and that the combination of several models was common among the systems with the highest performance.

However, the comparison between the systems does not make it possible to attribute the improvement to a single factor, because they differed in data sources, architectures and training methods. In addition, the study emphasizes the importance of calibrating model scores (Score Calibration), after some of the systems achieved high separability between real and fake speech, but weak performance on the actDCF metric.

The findings of the study emphasize that the evaluation of systems for detecting voice forgeries should be based not only on performance within a particular dataset, but also on examining the ability to generalize to speakers and forgery methods that did not appear in training. Accordingly, the present study will combine a comparison between approaches based on classical acoustic features and approaches based on pre-trained speech representations, using a controlled evaluation protocol and the EER metric. In addition, the possibility of incorporating external training data in order to improve generalization will be examined, while taking care to prevent overlap between the data sets.

Bibliographic Source

Wang, X., et al. (2024). ASVspoof 5: Crowdsourced Speech Data, Deepfakes, and Adversarial Attacks at Scale. arXiv:2408.08739. https://arxiv.org/abs/2408.08739